

Matching Theory (Game Theory)

Matching Theory is a branch of game theory and economics that studies **how to pair two groups of agents** based on their preferences, without using prices as the main mechanism.

The goal is to create a **stable and efficient matching**, where no two participants would both prefer to leave their assigned match and pair with each other instead.

Matching Theory is widely used in:

- School admissions
- College admissions
- Job recruitment
- Medical residency placement
- Marriage matching
- Organ donation
- Online dating platforms
- Housing allocation

Simple Definition

Matching Theory is the study of assigning individuals or organizations to one another according to their preferences in a way that produces stable and efficient outcomes.

Stable Matching

A matching is stable if there is no blocking pair.

Blocking Pair

A student and a college form a blocking pair if:

- the student prefers that college over their current assignment, and
- the college also prefers that student over one of its assigned students.

If such a pair exists, the matching is unstable because they would rather be matched together.

Gale–Shapley Algorithm (Deferred Acceptance Algorithm)

One of the most famous algorithms in Matching Theory.

Developed by:

- David Gale
- Lloyd Shapley

Steps

1. Every student proposes to their favorite college.

2. Colleges temporarily accept their preferred applicants.
3. Remaining students propose to their next choice.
4. Colleges keep the best applicants and reject the rest.
5. Continue until everyone is matched.

The algorithm always produces a stable matching.

Examples

Types of Matching Problems

1. One-to-One Matching

Example:

- Marriage
- Dating apps

One person is matched with one person.

2. Many-to-Many Matching

Example:

- Freelancers and projects
- Companies and suppliers

Both sides can have multiple matches.

3. Job Recruitment

Graduates rank companies.

Companies rank graduates.

The algorithm finds stable employment matches

Importance in Game Theory

Matching Theory solves problems where:

- Money cannot or should not determine the allocation.
- Participants have preferences over whom they are matched with.
- Stability is important to prevent participants from abandoning the assigned matching.

Advantages

- Produces stable outcomes.
 - Considers the preferences of both sides.
 - Reduces incentives for participants to deviate.
 - Can improve fairness in many allocation settings.
 - Has many practical applications.
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Limitations

- Participants may misreport preferences if the mechanism is not strategy-proof for them.
- Stability does not always maximize total satisfaction or social welfare.
- Large-scale matching can involve additional constraints (quotas, priorities, geographic restrictions).
- Real-world preferences may change over time.

Game Theory Perspective

Matching Theory differs from many strategic games because the main objective is finding a stable allocation rather than maximizing payoffs through repeated strategic interactions.

Each player:

- Has preferences.
- Chooses rankings.
- Is matched through a defined mechanism.
- Receives utility from the assigned match.

The equilibrium concept is stable matching, rather than a Nash equilibrium.

<u>Concept</u>	<u>Meaning</u>
Matching	Assigning participants to one another
Preferences	Ordered rankings of potential matches
Stable Matching	No blocking pair exists
Blocking Pair	Two participants who would both prefer to be matched together
Gale–Shapley Algorithm	Algorithm that guarantees a stable matching
One-to-One	One participant matched with one other participant

Concept

One-to-Many

Many-to-Many

Meaning

One participant matched with several others
(e.g., colleges)

Multiple matches allowed on both sides